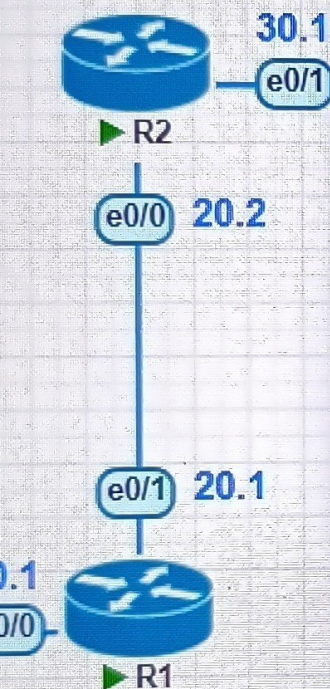
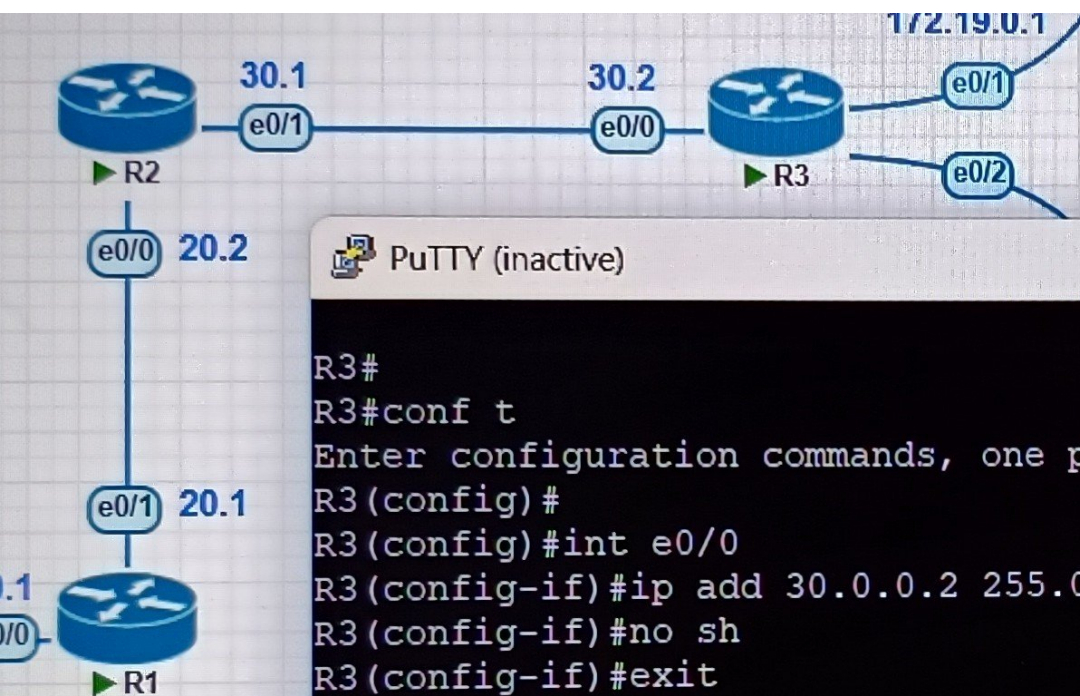


```
R1#
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#int e0/0
R1(config-if)#ip add 10.0.0.1 255.0.0.0
R1(config-if)#no sh
R1(config-if)#exit
R1(config)#
R1(config)#int e0/1
R1(config-if)#ip add 20.0.0.1 255.0.0.0
R1(config-if)#no sh
R1(config-if)#exit
R1(config)#
*Aug 23 11:01:27.414: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1(config)#exit
R1#
*Aug 23 11:01:55.820: %SYS-5-CONFIG_I: Configured from console by console
R1#write
Building configuration...
[OK]
R1#
```

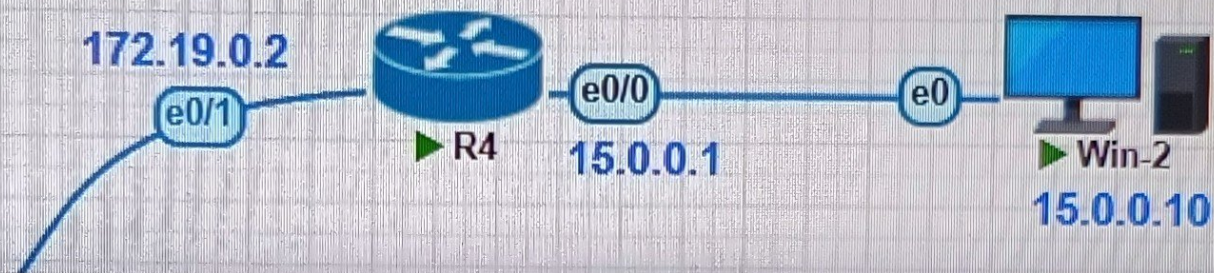
```
*Aug 23 10:59:03.642: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to down
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int e0/0
R2(config-if)#ip add 20.0.0.2 255.0.0.0
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#
R2(config)#int e0/1
R2(config-if)#ip add 30.0.0.1 255.0.0.0
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#write
      ^
% Invalid input detected at '^' marker.

R2(config)#exit
R2#
*Aug 23 11:02:22.230: %SYS-5-CONFIG_I: Configured from console by console
R2#write
Building configuration...
[OK]
R2#
```

PuTTY (inactive)

```
R3#
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#
R3(config)#int e0/0
R3(config-if)#ip add 30.0.0.2 255.0.0.0
R3(config-if)#no sh
R3(config-if)#exit
R3(config)#
R3(config)#int e0/1
R3(config-if)#ip add 172.19.0.1
*Aug 23 12:02:50.905: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R3(config-if)#ip add 172.19.0.1 255.255.0.0
R3(config-if)#no sh
R3(config-if)#exit
R3(config)#
R3(config)#int e0/2
R3(config-if)#ip add 21.0.0.1 255.0.0.0
R3(config-if)#no sh
R3(config-if)#exit
R3(config)#
```

PuTTY (inactive)

```
R4#
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#int e0/1
R4(config-if)#ip add 172.19.0.2 255.255.0.0
R4(config-if)#no sh
R4(config-if)#exit
R4(config)#
R4(config)#int e0/0
R4(config-if)#ip add 15.0.0.1 255.0.0.0
R4(config-if)#no sh
R4(config-if)#exit
R4(config)#
*Aug 23 12:09:36.907: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R4(config)#exit
R4#
*Aug 23 12:11:22.628: %SYS-5-CONFIG_I: Configured from console by console
R4#show ip int brief
```

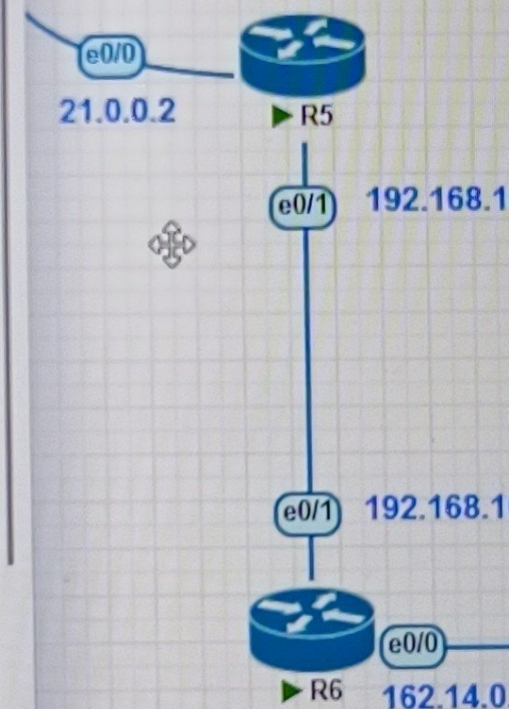


```

R5#
R5#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R5(config)#int e0/0
R5(config-if)#ip add 21.0.0.2 255.0.0.0
R5(config-if)#no sh
R5(config-if)#exit
R5(config)#
R5(config)#int e0/1
R5(config-if)#ip add 192.168.168.1 255.255.255.0
R5(config-if)#no sh
R5(config-if)#exit
R5(config)#
*Aug 23 12:10:03.006: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R5(config)#exit
R5#s
*Aug 23 12:11:04.848: %SYS-5-CONFIG_I: Configured from console by console
R5#show ip int brief

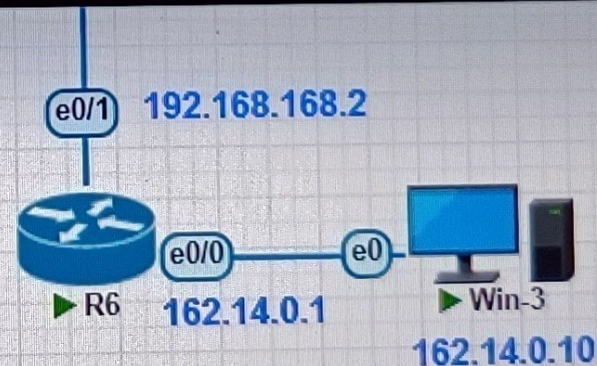
```

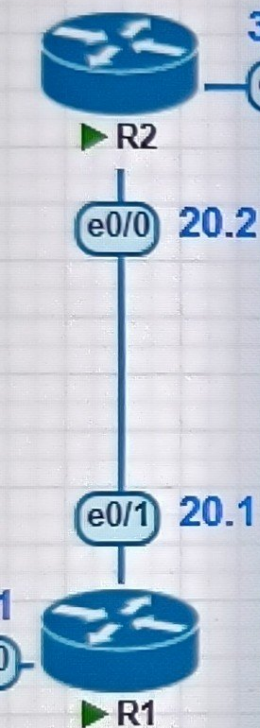
Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	21.0.0.2	YES	manual	up	



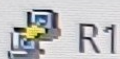

```
R6#
R6#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R6(config)#int e0/1
R6(config-if)#ip add 192.168.168.2 255.25.255.0
Bad mask 0xFF19FF00 for address 192.168.168.2
R6(config-if)#ip add 192.168.168.2 255.255.255.0
R6(config-if)#no sh
R6(config-if)#exit
R6(config)#
R6(config)#int e0/0
R6(config-if)#ip add 162
*Aug 23 12:09:46.015: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R6(config-if)#ip add 162.14.0.1 255.255.0.0
R6(config-if)#no sh
R6(config-if)#exit
R6(config)#
R6(config)#exit
R6#
*Aug 23 12:10:34.588: %SYS-5-CONFIG_I: Configured from console by console
R6#show ip int brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
-----------	------------	-----	--------	--------	----------





```
R1#  
3 R1#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
R1(config)#  
R1(config)#ip route 30.0.0.1 255.0.0.0 20.0.0.2  
%Inconsistent address and mask  
R1(config)#ip route 30.0.0.0 255.0.0.0 20.0.0.2  
R1(config)#ip route 172.19.0.0 255.255.0.0 20.0.0.2  
R1(config)#ip route 15.0.0.0 255.0.0.0 20.0.0.2  
R1(config)#ip route 21.0.0.0 255.0.0.0 20.0.0.2  
R1(config)#ip route 192.168.168.0 255.255.255.0 20.0.0.2  
R1(config)#ip route 162.14.0.0 255.255.0.0 20.0.0.2  
R1(config)#  
R1(config)#
```

R1

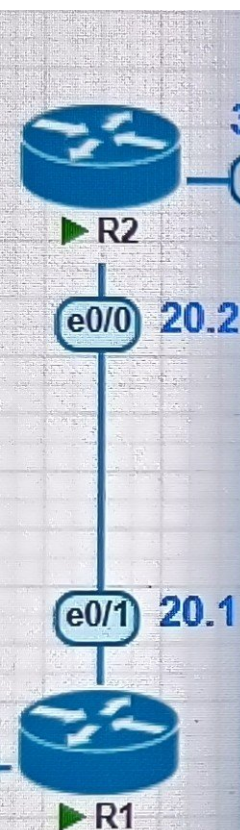
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.0.0.0/8 is directly connected, Ethernet0/0
L 10.0.0.1/32 is directly connected, Ethernet0/0
S 15.0.0.0/8 [1/0] via 20.0.0.2
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 20.0.0.0/8 is directly connected, Ethernet0/1
L 20.0.0.1/32 is directly connected, Ethernet0/1
S 21.0.0.0/8 [1/0] via 20.0.0.2
S 30.0.0.0/8 [1/0] via 20.0.0.2
S 162.14.0.0/16 [1/0] via 20.0.0.2
S 172.19.0.0/16 [1/0] via 20.0.0.2
S 192.168.168.0/24 [1/0] via 20.0.0.2

R1#

R1#



R2#

R2#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R2(config)#

R2(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1

R2(config)#ip route 172.19.0.0 255.255.0.0 30.0.0.2

R2(config)#ip route 15.0.0.0 255.0.0.0 30.0.0.2

R2(config)#ip route 21.0.0.0 255.0.0.0 30.0.0.2

R2(config)#ip route 192.168.168.0 255.255.255.0 30.0.0.2

R2(config)#ip route 162.14.0.0 255.255.0.0 30.0.0.2

R2(config)#

R2(config)#

R2(config)#

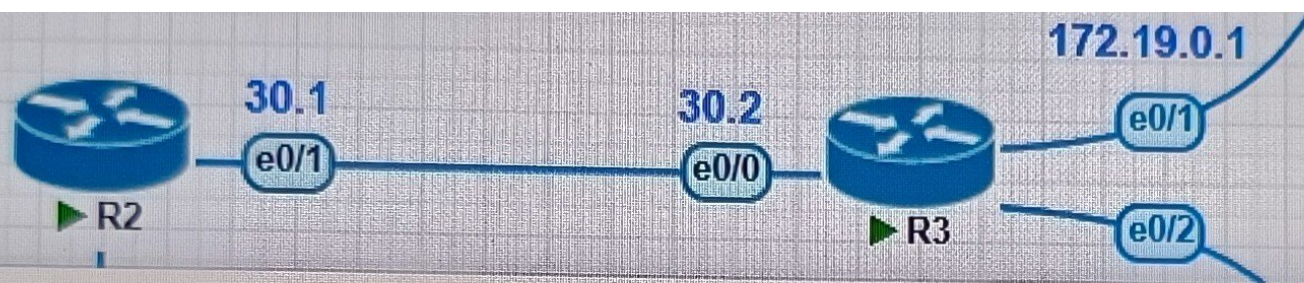
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 20.0.0.1
S 15.0.0.0/8 [1/0] via 30.0.0.2
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 20.0.0.0/8 is directly connected, Ethernet0/0
L 20.0.0.2/32 is directly connected, Ethernet0/0
S 21.0.0.0/8 [1/0] via 30.0.0.2
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 30.0.0.0/8 is directly connected, Ethernet0/1
L 30.0.0.1/32 is directly connected, Ethernet0/1
S 162.14.0.0/16 [1/0] via 30.0.0.2
S 172.19.0.0/16 [1/0] via 30.0.0.2
S 192.168.168.0/24 [1/0] via 30.0.0.2

R2#

R2#



PuTTY (inactive)

```
R3#
R3#
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip route 10.0.0. 255.0.0.0 30.0.0.1
10.
e0% Invalid input detected at '^' marker.

R3(config)#ip route 10.0.0.0 255.0.0.0 30.0.0.1
R3(config)#ip route 20.0.0.0 255.0.0.0 30.0.0.1
R3(config)#ip route 15.0.0.0 255.0.0.0 172.19.0.2
R3(config)#ip route 192.168.168.0 255.255.255.0 21.0.0.2
R3(config)#ip route 168.14.0.0 255.255.0.0 21.0.0.2
R3(config)#
R3(config)#
R3(config)#
R3(config)#
```



```
R3#conf t
```

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

```
R3(config)#no ip route 168.14.0.0 255.255.0.0 21.0.0.2
```

```
R3(config)#ip route 162.14.0.0 255.255.0.0 21.0.0.2
```

```
R3(config)#
```

```
R3(config)#exit
```


E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

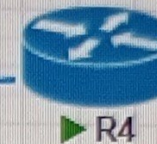
Gateway of last resort is not set

```
R3#
S    10.0.0.0/8 [1/0] via 30.0.0.1
S    15.0.0.0/8 [1/0] via 172.19.0.2
S    20.0.0.0/8 [1/0] via 30.0.0.1
    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        21.0.0.0/8 is directly connected, Ethernet0/2
L        21.0.0.1/32 is directly connected, Ethernet0/2
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        30.0.0.0/8 is directly connected, Ethernet0/0
L        30.0.0.2/32 is directly connected, Ethernet0/0
S    162.14.0.0/16 [1/0] via 21.0.0.2
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.19.0.0/16 is directly connected, Ethernet0/1
L        172.19.0.1/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 21.0.0.2
```

R3#

172.19.0.2

e0/1



e0/0

15.0.0.1

PuTTY (inactive)

```
R4#
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#ip route 10.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 20.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 30.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 21.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 192.168.168.0 255.255.255.0 172.19.0.1
R4(config)#ip route 162.14.0.0 255.255.0.0 172.19.0.1
R4(config)#
R4(config)#
R4(config)#
```


N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

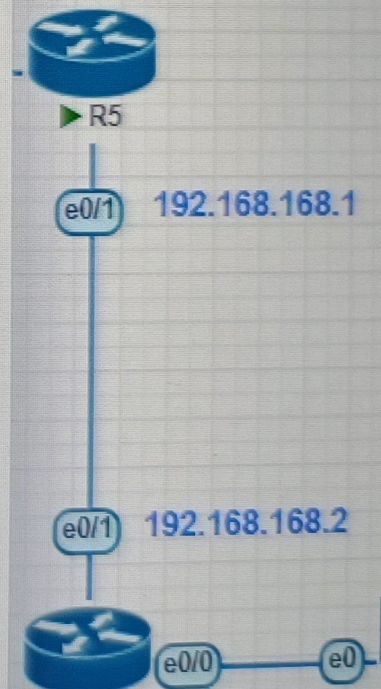
Gateway of last resort is not set

```
S    10.0.0.0/8 [1/0] via 172.19.0.1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        15.0.0.0/8 is directly connected, Ethernet0/0
L        15.0.0.1/32 is directly connected, Ethernet0/0
S    20.0.0.0/8 [1/0] via 172.19.0.1
S    21.0.0.0/8 [1/0] via 172.19.0.1
S    30.0.0.0/8 [1/0] via 172.19.0.1
S    162.14.0.0/16 [1/0] via 172.19.0.1
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.19.0.0/16 is directly connected, Ethernet0/1
L        172.19.0.2/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 172.19.0.1
```

R4#

R4#


```
R5#  
R5#  
R5#  
R5#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
R5(config)#ip route 10.0.0.0 255.0.0.0 21.0.0.1  
R5(config)#ip route 20.0.0.0 255.0.0.0 21.0.0.1  
R5(config)#ip route 30.0.0.0 255.0.0.0 21.0.0.1  
R5(config)#ip route 15.0.0.0 255.0.0.0 21.0.0.1  
R5(config)#ip route 172.19.0.0 255.255.0.0 21.0.0.1  
R5(config)#ip route 162.14.0.0 255.255.0.0 192.168.168.2  
R5(config)#  
R5(config)#  
R5(config)#
```



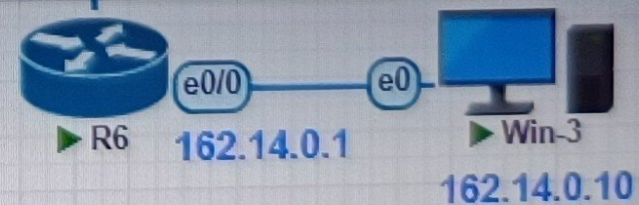
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S 10.0.0.0/8 [1/0] via 21.0.0.1
S 15.0.0.0/8 [1/0] via 21.0.0.1
S 20.0.0.0/8 [1/0] via 21.0.0.1
  21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    21.0.0.0/8 is directly connected, Ethernet0/0
L    21.0.0.2/32 is directly connected, Ethernet0/0
S 30.0.0.0/8 [1/0] via 21.0.0.1
S 162.14.0.0/16 [1/0] via 192.168.168.2
S 172.19.0.0/16 [1/0] via 21.0.0.1
  192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.168.0/24 is directly connected, Ethernet0/1
L    192.168.168.1/32 is directly connected, Ethernet0/1
```

R5#


```
R6#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R6(config)#
R6(config)#ip route 10.0.0.0 255.0.0.0 192.168.168.1
R6(config)#ip route 20.0.0.0 255.0.0.0 192.168.168.1
R6(config)#ip route 30.0.0.0 255.0.0.0 192.168.168.1
R6(config)#ip route 15.0.0.0 255.0.0.0 192.168.168.1
R6(config)#ip route 21.0.0.0 255.0.0.0 192.168.168.1
R6(config)#ip route 172.19.0.0 255.255.0.0 192.168.168.1
R6(config)#
```



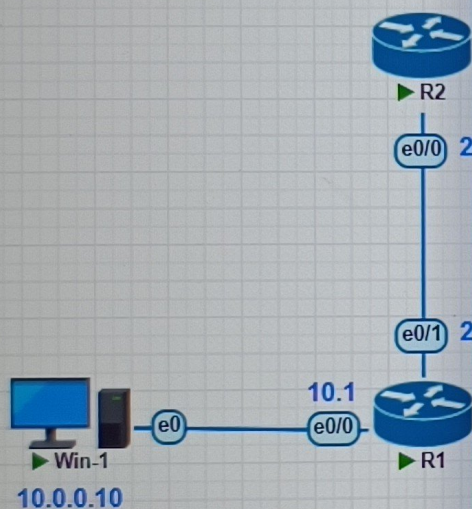
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    10.0.0.0/8 [1/0] via 192.168.168.1
S    15.0.0.0/8 [1/0] via 192.168.168.1
S    20.0.0.0/8 [1/0] via 192.168.168.1
S    21.0.0.0/8 [1/0] via 192.168.168.1
S    30.0.0.0/8 [1/0] via 192.168.168.1
    162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        162.14.0.0/16 is directly connected, Ethernet0/0
L        162.14.0.1/32 is directly connected, Ethernet0/0
S    172.19.0.0/16 [1/0] via 192.168.168.1
    192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.168.0/24 is directly connected, Ethernet0/1
L        192.168.168.2/32 is directly connected, Ethernet0/1
```

R6#

R6#



QEMU (Win-1)



My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-I...

Old Java ASD

PuTTY

Tftpd32

Network Connections

File Edit View Favorites Tools Advanced Help

Back Forward Stop Search Folders View

Address Network Connections

Local Area Connection Properties

General Advanced

Internet Protocol (TCP/IP) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 10 . 0 . 0 . 10

Subnet mask: 255 . 0 . 0 . 0

Default gateway: 10 . 0 . 0 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server:

Alternate DNS server:

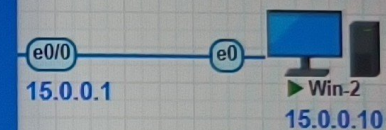
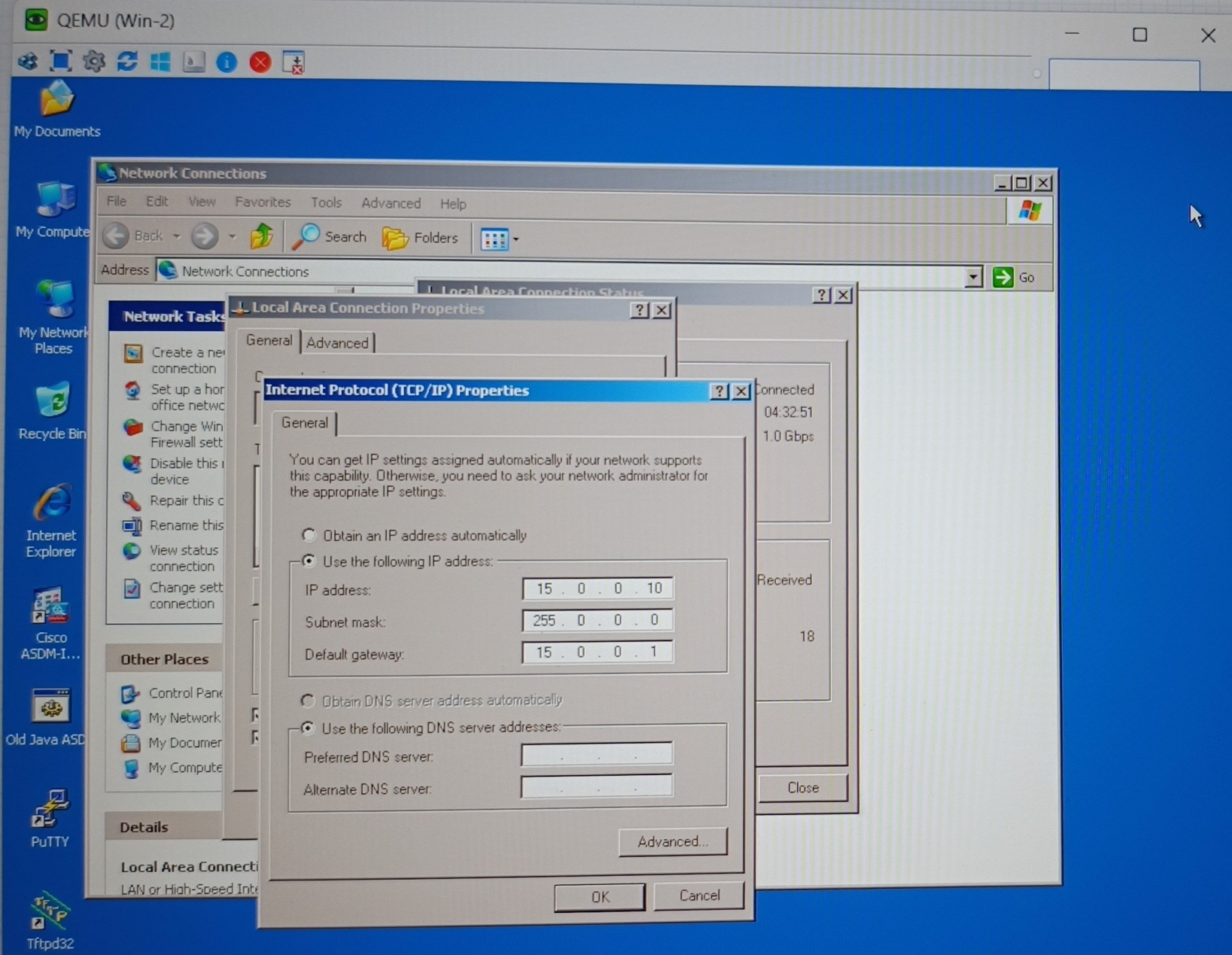
Advanced

OK Cancel

Connected 04:32:30 1.0 Gbps

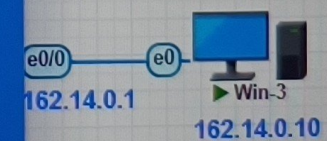
Received 33

Close



92.168.168.1

92.168.168.2



QEMU (Win-3)

My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-I...

Old Java ASD

PuTTY

Tftpd32

Local Area Connection Properties

General Advanced

Internet Protocol (TCP/IP) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 162 . 14 . 0 . 10

Subnet mask: 255 . 255 . 0 . 0

Default gateway: 162 . 14 . 0 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

Advanced...

OK

Cancel

Connected

04:34:16

1.0 Gbps

Received

99

18

Close

Details

Local Area Connection

LAN or High-Speed Internet

Network Connections

Local Area Connection...

Local Area Connecti...

C:\WINDOWS\system...

4:33 PM

e0/0
15.0.0.1

Win-2
15.0.0.10

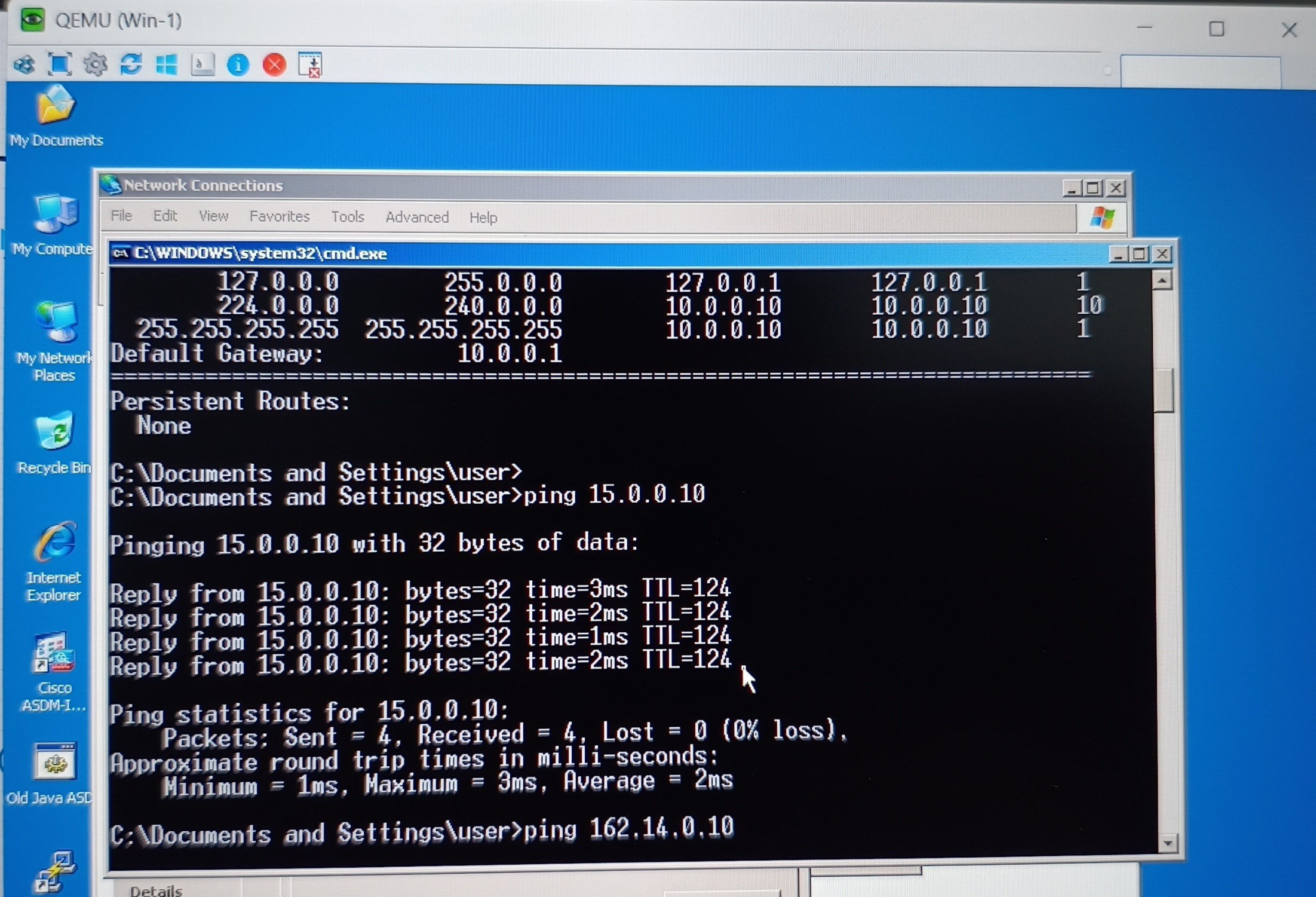
192.168.168.1

192.168.168.2

e0/0
162.14.0.1

Win-3
162.14.0.10

Physical





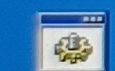
My Documents



My Computer

My Network
Places

Recycle Bin

Internet
ExplorerCisco
ASDM-I...

Old Java ASD



PUTTY

Network Connections

File Edit View Favorites Tools Advanced Help

C:\WINDOWS\system32\cmd.exe

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=4ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
      Minimum = 2ms, Maximum = 4ms, Average = 3ms

C:\Documents and Settings\user>
```

Details

Advanced...



My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-I...

Old Java ASDM

PuTTY

```
C:\Documents and Settings\user>ping 162.14.0.10
```

```
Pinging 162.14.0.10 with 32 bytes of data:
```

```
Reply from 162.14.0.10: bytes=32 time=3ms TTL=124
Reply from 162.14.0.10: bytes=32 time=1ms TTL=124
Reply from 162.14.0.10: bytes=32 time=1ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
```

```
Ping statistics for 162.14.0.10:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms
```

```
C:\Documents and Settings\user>ping 10.0.0.10
```

```
Pinging 10.0.0.10 with 32 bytes of data:
```

```
Reply from 10.0.0.10: bytes=32 time=4ms TTL=124
Reply from 10.0.0.10: bytes=32 time=1ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
```

```
Ping statistics for 10.0.0.10:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms
```

```
C:\Documents and Settings\user>
```


WORK.COM
31168
Version ...

kbit/s

es Received

0

SUREN

PLAY

QEMU (Win-3)

C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=2ms TTL=123

Reply from 10.0.0.10: bytes=32 time=3ms TTL=123

Reply from 10.0.0.10: bytes=32 time=3ms TTL=123

Reply from 10.0.0.10: bytes=32 time=3ms TTL=123

Ping statistics for 10.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Reply from 15.0.0.10: bytes=32 time=3ms TTL=124

Ping statistics for 15.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>



Win-1

10.0.0.10